Nevin John Selby

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Education

University of Wisconsin - Madison: Masters in Data Science | GPA: 3.70/4.0 Sept 2023 – May 2025

- **Coursework:** Machine Learning Systems, Deep Learning, Statistical Models, MLOps, Distributed Systems

Indian Institute of Information Technology: Bachelors in Computer Science | GPA: 9.15/10.0 Aug 2019 – May 2023

- **Coursework:** Machine Learning, Data Structures, Cloud Computing, Statistical Learning

Experience

Data Science Intern, Wisconsin School of Business – Madison, WI Sept 2024 – Present

- Architected an end-to-end NLP pipeline using fine-tuned GPT-3.5-Turbo for sentiment analysis on 12GB customer feedback, achieving 83% accuracy and enabling real-time customer experience insights
- Developed scalable ETL workflows using advanced SQL, reducing query latency by 40% while implementing robust data quality checks
- Designed a custom rate-limiting and checkpointing system using Redis, reducing API costs by 25% and enabling processing of 1M+ daily requests
- Established MLOps best practices including CI/CD pipelines, model versioning, and A/B testing, resulting in 18% improvement in model performance across all categories

Research Assistant, UW College of Agricultural & Life Sciences – Madison, WI Sept 2024 – Present

- Optimized a computer vision pipeline using PyTorch and ResNet, improving cranberry phenology classification accuracy by 25% through transfer learning and semi-supervised techniques
- Spearheaded serverless ML infrastructure on AWS (Lambda, S3, SageMaker) with automated model retraining, reducing processing time by 40% and enabling real-time inference
- Enhanced model performance through custom data augmentation pipeline using albumentations library, increasing YOLOv5 model precision by 15% through intelligent sampling
- Streamlined annotation workflows using CVAT and active learning, reducing manual labeling effort by 30% across 12,000+ images

Graduate Researcher, Wisconsin Institute for Discovery – Madison, WI Jan 2024 – Jun 2024

- Innovated label-efficient learning strategies using foundation models and synthetic data generation, reducing annotation costs by 80%
- Leveraged semi-supervised learning pipeline integrating weak supervision and active learning, automating 90% of labeling workflows
- Deployed comprehensive model monitoring system using MLflow and Weights & Biases, enabling real-time performance tracking and automatic retraining triggers

Projects

Large Language Model Fine-tuning Framework [GitHub] Jan 2024 - Jun 2024

- Modeled distributed training framework using DeepSpeed and PyTorch Lightning, enabling efficient fine-tuning of 6B+ parameter models with 40% reduced computing costs
- Integrated parameter-efficient fine-tuning techniques (LoRA, QLoRA) achieving 93.5% F1-score on GLUE benchmark while reducing memory footprint by 60%

MLOps Pipeline Builder Jan 2023 - May 2023

- Designed zero-code ML platform using Streamlit and MLflow, supporting automated feature engineering, model validation, and A/B testing
- Validated scalable data processing using Dask and Ray, enabling parallel processing of 10M+ rows while maintaining sub-second prediction latency

Skills

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, LLMs, Computer Vision, NLP

MLOps: Docker, Kubernetes, MLflow, Weights & Biases, Model Registry, Feature Store

Cloud and Data: AWS (SageMaker, Lambda, S3), Snowflake, PostgreSQL

Programming: Python, SQL, C++, JavaScript